Tobias Mwangi-Hartley

Data Scientist | Machine Learning

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PROFESSIONAL SUMMARY

Data scientist with six years of experience in analytics and data science, including more than four years as a data scientist. Builds and operationalises forecasting and churn propensity models in Python and SQL, including feature engineering, scheduled batch scoring and automated testing. Strong statistical background and a track record of translating analysis into operational decisions. Moving towards machine learning engineering.

TECHNICAL SKILLS

Python: pandas, scikit-learn, LightGBM, XGBoost, statsmodels, pytest, Streamlit; PyTorch, torchvision (course project; CNN fine-tuning)

Cloud and data: GCP (BigQuery, Vertex AI Workbench, Cloud Run), SQL, dbt, Looker

Engineering tools: Git, Docker (Dockerfile for Streamlit app), Jupyter

Testing and deployment: pytest, Docker, Streamlit, scheduled notebook workflows, dbt Cloud scheduler

Methods: feature engineering, model evaluation, time-series forecasting, A/B testing, causal inference, Bayesian statistics

R: tidyverse, brms, R Markdown

PROFESSIONAL EXPERIENCE

Data Scientist | Thistlebox, Edinburgh | Feb 2022–Present

- Built a LightGBM demand forecasting model for weekly orders by recipe and region, six weeks ahead. Engineered calendar, price and menu-position features; reduced WAPE from 23% to 14% and food waste by an estimated 9%.
- Operationalised the LightGBM forecasting workflow as a scheduled Vertex AI Workbench notebook writing results to BigQuery, with outputs consumed by operations in Looker.
- Started writing pytest unit tests for forecasting feature code in 2025, reaching approximately 60% coverage.
- Built an XGBoost churn propensity model scored weekly in BigQuery to target retention offers. A 10% holdout showed churn was 2.1 percentage points lower in the treated group.
- Wrote production SQL in BigQuery and maintained dbt models and scheduled data workflows for the analytics layer.
- Designed and analysed approximately 40 A/B tests per year for product and growth teams. Built a Streamlit sequential testing calculator used by approximately 25 colleagues; wrote its Dockerfile, while the data platform team handled deployment.
- Wrote the Dockerfile for a Streamlit sequential-testing application used by approximately 25 colleagues; the data platform team deployed it on Cloud Run.
- Presented results to leadership monthly and delivered internal “Stats for PMs” training three times.

Junior Data Analyst | Firth Energy Retail, Glasgow | Sep 2020–Jan 2022

- Conducted customer segmentation and tariff analysis in R and SQL.
- Automated monthly regulator reports using R Markdown, saving the team approximately two days per month.

PROJECT

Image Classification Course Project | 2025

- Fine-tuned a small convolutional neural network in PyTorch on a public food images dataset, achieving approximately 80% accuracy. Completed as a learning project using torchvision and Google Colab.

EDUCATION

MSc Statistics, Distinction | University of the Lothians | 2020
Dissertation: Bayesian hierarchical models for small-area energy demand

BSc Mathematics | University of the Lothians | 2019

LANGUAGES

English (native); Swahili (conversational)